Bhinu Puvvala

+1 (312)-918-4861 | bhinusantosh@gmail.com | linkedin.com/in/bhinu-santosh | github.com/bhinu

Education

University of Wisconsin - Madison Expected Graduation: May 2027

Bachelor of Science in Computer Science & Data Science; Minor: Economic Analytics *GPA: 3.74/4.0 (Dean's List)*

- **Relevant Coursework:** Big Data Systems, Computer Vision, Statistics, Data Structures & Algorithms, Machine Organization, OS

Experience

Software Engineering Intern May 2025 – Aug 2025

Methix (Microsoft-backed) *New York, NY*

- Built and deployed Python/Flask backend services supporting real-time decision workflows for 5,000+ weekly users, reducing p95 response latency by 30% through caching and optimized data access
- Improved database query performance by 52% (250ms→120ms) by redesigning indexes and access patterns, enabling reliable handling of 2× peak concurrent traffic
- Increased production reliability by implementing structured logging, latency tracing, and automated health checks, reducing incident detection time and improving system stability

Undergraduate Researcher Jan 2025 – May 2025

GLUE Lab, University of Wisconsin - Madison *Madison, WI*

- Developed large-scale perception pipelines over 10TB+ multispectral satellite data, improving scene-level classification accuracy by 20% through feature extraction and model calibration
- Reduced false positives by 53% and end-to-end inference latency by 86% by designing distributed, fault-tolerant processing workflows using Dask and GeoPandas
- Validated model robustness by running systematic evaluations and ablation studies across regions and sensor conditions, improving generalization in data-scarce environments

Data Analyst Intern June 2024 – Aug 2024

Diyar United Company *Kuwait City, Kuwait*

- Scaled data ingestion by automating Python/SQL ETL pipelines processing 500K+ records daily, increasing throughput by 3× and saving 15+ engineering hours per week
- Enabled data-driven decision-making for 10K+ clients by designing a normalized star-schema warehouse and analytical dashboards for experimentation and segmentation

Undergraduate Teaching Assistant Sept 2025 – Present

University of Wisconsin - Madison *Madison, WI*

- Led weekly analytical problem-solving sessions for 60+ students in Linear Algebra and Pre-Calculus, improving average performance by 15% through structured quantitative reasoning

Event Manager Aug 2024 – Present

Wisconsin Union, University of Wisconsin - Madison *Madison, WI*

- Led end-to-end logistics for campus events annually by managing and coordinating a 30+ person team, implementing digital scheduling and process optimizations that reduced setup time and coordination delays for large-scale events

Projects

WhismurAI | *React, FastAPI, Web Audio API, Deepgram, OpenAI, Fish Audio*

- Built a real-time, low-latency audio perception and translation pipeline supporting 20+ languages by streaming raw audio through STT, semantic translation, and TTS under strict end-to-end latency constraints
- Architected a FastAPI-based streaming backend for real-time media processing, preserving speaker identity and achieving sub-second end-to-end playback latency to improve user experience

Gesture: Real-Time Human-Computer Interaction | *TensorFlow, MediaPipe, OpenCV, Electron.js*

- Achieved 98% gesture recognition accuracy and 30ms real-time latency by optimizing MediaPipe pipelines with custom TensorFlow models and OpenCV preprocessing to enable robust control in challenging conditions
- Reduced command execution time by 40% through the engineering of a modular recognition framework and Electron.js calibration interface supporting 12+ user-defined commands integrated with productivity tools

HackSherpa | *Next.js, React, TypeScript, Supabase, Flask*

- Developed and deployed a scalable full-stack system, processing daily API requests with p95 latency under 200ms using NLP-driven recommendations

Technical Skills

Languages: Python, C/C++, Java, SQL, JavaScript, TypeScript, Bash
Perception & ML: PyTorch, TensorFlow, Statistical Modeling, Feature Engineering, Sensor Data Processing
Data & Compute: Pandas, NumPy, Dask, GeoPandas
Systems & Infrastructure: Docker, Redis, PostgreSQL, MySQL, MongoDB, Kubernetes, Git